Agent Factory Tycoon — Game Concept Document

Based on AgentFactory (Panaversity) — agentfactory.panaversity.org

Prepared: 20 June 2026

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Agent Factory Tycoon

Build, Train & Run Your Own AI Workforce — One Digital Worker at a Time

An educational game concept that teaches Agentic AI, Digital FTEs, and Spec-Driven Development through play.

Based on the AgentFactory live book — https://agentfactory.panaversity.org/docs

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Executive Summary

Agent Factory Tycoon is a cozy, mobile-first build-and-manage game that teaches real Agentic AI skills through play. The player runs a tiny startup and grows it by **building, training, and deploying Digital FTEs** — AI workers assembled from four parts (**Brain, Hands, Training, Shift**) — then watches them run the business 24/7. Without ever reading a manual, players absorb the core ideas of the AgentFactory curriculum: the Digital FTE, Spec-Driven Development, and the 10-80-10 human-in-the-loop rhythm.

The problem it solves. Agentic AI is one of the most valuable skills of the decade, but the on-ramp is intimidating — walls of theory, jargon, and setup. Agent Factory Tycoon removes that wall, turning the first hour of learning into something genuinely fun.

How it plays. A tight 60–90 second loop: a customer appears → you build a Digital FTE by dragging four parts together → you write its job on a fill-in-the-blank "Spec Card" → you press GO and watch it work → you verify the result → you earn coins, XP, and unlocks. Twenty-six short levels mirror AgentFactory's 26 real crash courses, so the game doubles as an on-ramp to the actual curriculum.

Why it works. The design is grounded in proven frameworks, not guesswork:

- **Fun** — Lazzaro's 4 Keys to Fun, Flow theory, and heavy "game feel" juice.
- **Easy** — one new concept per level, "show don't tell" onboarding, no punishing fail states.
- **Interactive** — direct-manipulation drag/drop, meaningful build choices, a living reactive world.
- **Beautiful** — a stylized 2.5D low-poly art direction (Unity + URP) that ships at 60fps on mid-range phones and in-browser.

The hook. A friendly mascot named **Nova** guides every step, and the signature "silly spec" gag teaches the hardest lesson — clear specifications — through laughter instead of lectures. Approved workers keep earning coins even while the player is offline, embodying the Digital FTE promise and driving retention.

Target audience. Total beginners, students, and professionals who want to understand Digital FTEs in 30 minutes of play.

Status & ask. This document is a complete game-design blueprint: concept, full 26-level map, four-pillar design spec, end-to-end Level 1 storyboards, and a character bio. The recommended next step is a small **MVP** proving the assemble → spec → verify loop; if Level 1 hits >90% completion and testers smile at the core loop, the concept is validated and ready to scale.

Part 1 — Concept Paper

1. What Is a Concept Paper?

A concept paper is a short design blueprint that defines a game **before** building it: its idea, who it's for, what the player learns, how it plays, how it looks, and why people will love it. It is the "pitch + plan" that aligns everyone before a single line of code is written. This document is that blueprint.

2. The Core Idea (One Sentence)

> You run a tiny startup. To grow, you **build, train, and deploy Digital FTEs** (AI workers) by snapping together their **Brain, Hands, Training, and Shift** — and watch them run your business 24/7.

It is a **cozy build-and-manage game** (think *Stardew Valley* meets *Two Point Hospital*) where, without realizing it, the player learns real Agentic AI concepts.

3. Target Audience

Player	Why they'll play
Total beginners	Learn AI agents through play, zero jargon
Students	Fun on-ramp before the real crash courses
Professionals	Understand Digital FTEs in 30 minutes

Difficulty: deliberately easy. Drag, drop, tap. No reading walls, no harsh fail-states — only "your agent could do better, try again."

4. Learning Objectives (Hidden Inside the Fun)

Each game mechanic secretly maps to a real AgentFactory concept:

In the game you...	You're really learning...
Assemble a worker from 4 parts	The Digital FTE anatomy: Brain, Hands, Training, Shift

In the game you...	You're really learning...
Write a "job order" card	Spec-Driven Development (clear specs = good results)
Pick which model powers a worker	Choosing an engine (Claude / GPT / Gemini)
Plug in tool-chips	MCP & tools ("Hands")
Give a worker a skill-book	Agent Skills ("Training")
Set intent, watch, approve	The 10-80-10 rule
Manage a growing team	The Seven Invariants of an AI-native company

5. Core Gameplay Loop (Easy & Addictive)

```
   1. A CUSTOMER appears with a request
   2. BUILD a Digital FTE (drag 4 parts)
   3. WRITE the job spec (fill-in-the-blank)
   4. PRESS GO -> agent works (cute animation)
   5. VERIFY the result -> approve / improve
   6. EARN coins + XP -> unlock new parts
   ... repeat, scale up ...
```

A full loop takes **60–90 seconds** — perfect for quick, satisfying sessions.

6. Game Mechanics That Make It Fun & Interactive

- **Drag-and-Drop Worker Builder** — Snap together glowing puzzle pieces: a **Brain** orb, **Hand** tools, a **Training** book, and a **Shift** clock. Pieces "click" with a satisfying sound and sparkle.
- **The Spec Card** — A fill-in-the-blank card: *"I want a worker that ___ when ___."* Vague answers make the agent do something silly (funny!); clear answers make it nail the task. Players *feel* why specs matter.
- **Live "Work" Animation** — After hitting GO, your little agent bustles around — typing, sorting mail, answering chats — with progress bars and emoji reactions.
- **Verify Mini-Game** — A quick "thumbs up / fix it" choice teaches that humans verify, agents execute.
- **Factory Builder / Idle Layer** — Approved workers keep earning coins even when you are offline (idle-game retention hook).
- **Progression & Unlocks** — 26 short levels mirror the 26 crash courses. Each unlock is a new part, tool, or worker type.

7. Visual & Audio Design (Good Graphics)

- **Art style:** clean, cheerful 3D isometric or polished 2.5D cartoon — bright, rounded, friendly (Two Point / Mini Metro charm). Reads great on mobile.
- **Characters:** adorable robot/agent mascots with expressive faces and idle animations.

- **UI:** large tap targets, minimal text, icon-first, soft shadows, smooth transitions.
- **Color:** energetic gradients (purple/teal "tech" palette) tied to AgentFactory branding.
- **Audio:** satisfying click/ding feedback, upbeat low-fi background music, celebratory chimes on success.
- **Juice:** screen-shake on big wins, confetti on level-up, particle sparkles on assembly.

8. Extra Functionalities (Powerful & Interactive)

- Leaderboards & daily challenges ("Build the best Support FTE today")
- Nova, the tutorial mascot — a friendly guide who helps for the first 3 levels, then steps back
- Themed campaigns — Sales city, Healthcare hospital, Legal tower
- Co-op / share — share your factory layout, compare agents with friends
- Accessibility — colorblind mode, one-hand play, adjustable text size
- "Real concept" cards — 1-tap link connecting each mechanic to the actual AgentFactory course
- Gentle notifications — "Your night-shift FTE earned 200 coins!"

9. Suggested Tech Stack

- **Engine:** Unity (best 2.5D/3D + mobile + web export) or Godot (free, lightweight)
- **Web build:** WebGL so it runs in-browser, matching AgentFactory's no-install labs
- **Platforms:** Web first, then iOS/Android
- **Backend (optional):** lightweight save/leaderboard service

10. MVP Scope (What to Build First)

1. One customer type + one worker to build
2. The 4-part drag-and-drop builder
3. The Spec Card mechanic
4. The work animation + verify step
5. Coins, XP, and 5 levels

If players smile at the **assembly + spec + verify** loop, the concept works — then scale up.

11. Why This Works

- **Easy:** drag, drop, tap — no manuals.
- **Fun:** cute agents, idle earnings, unlocks, leaderboards.
- **Good graphics:** bright, polished, mobile-friendly isometric world.
- **Interactive & powerful:** every tap teaches a real Agentic AI concept usable later in AgentFactory's actual courses.
- **Builds intuition** so the real 15-hour Digital FTE track feels familiar, not scary.

Part 2 — Level 1 Walkthrough: "Your First Hire"

Goal: Build and deploy your very first Digital FTE — a Support Bot. **Duration:** ~3 minutes · **Difficulty:** Tutorial (impossible to lose)

Scene 0 — Opening (Cinematic, 8 seconds)

Visual: Camera pans over a tiny, empty startup office — one desk, a flickering "OPEN" sign, dust motes in sunlight. Upbeat low-fi music starts.

Nova — your friendly guide mascot — pops in. Nova is a small, hovering orb of light that glows in the game's signature violet-and-teal, with two big expressive eyes and a wisp of a "spark tail." She is upbeat, a little cheeky, and endlessly encouraging — half mentor, half hype-friend. (Nova appears throughout the game as the player's companion and tutor, celebrating wins, cracking gentle jokes, and nudging with just-in-time hints — never lecturing.)

> "Welcome, boss! This is YOUR company. Problem is... you're the only one here, and the customers are already lining up. Let's hire some help — the AI kind. Tap anywhere to begin!"

Player action: Tap screen.

Concept taught: A company's value comes from its *workforce*, not just its tools. (Thesis: Digital FTEs are the unit of business value.)

Scene 1 — A Customer Arrives

Customer (Mia): "Hi! I bought a backpack but it arrived with a broken zipper. Can someone help me return it?"

Nova: "A customer needs help! You COULD answer this yourself... every single time... forever. Or — we build a worker who handles it 24/7. Let's build one!"

Player action: Tap **[Build a Worker]**.

Concept taught: The difference between *you doing the work* vs. *manufacturing a worker who does it* (Mode 1 vs. Mode 2).

Scene 2 — The Worker Builder (Core Mechanic)

A clean assembly bench appears: a glowing empty robot silhouette with 4 empty slots, and 4 trays of puzzle pieces.

Part 1 — The BRAIN: Drag the purple Brain orb into the head slot (click + sparkle; eyes light up). *Concept: Brain = the reasoning model (Claude/GPT/Gemini).*

Part 2 — The HANDS: Drag the Orders and Chat tool-chips into the hand slots. *Concept: Hands = MCP + tools that let an agent act on real systems.*

Part 3 — The TRAINING: Drag the "Returns Policy" skill-book; the robot reads it. *Concept: Training = Agent Skills (domain knowledge + SOPs).*

Part 4 — The SHIFT: Drag the 24/7 clock; the robot powers up and waves (confetti). *Concept: Shift = autonomous 24/7 operation. "A tool waits for a prompt; a Digital FTE runs on its own."*

Scene 3 — The Spec Card (The Clever Teaching Moment)

A "Job Order" card slides up with fill-in-the-blank slots and tappable word-tiles.

Card: *"My worker should [____] when [____], then [____]."*

Trick — try the silly path first: tapping "panic" + "eat lunch" makes the robot spin in circles and throw papers (sad trombone). Nova laughs: "See? Unclear instructions = chaos. Let's give it a REAL spec."

Correct path: *"My worker should **help the customer** when **a return request arrives**, then **confirm the refund**."* The card glows green.

Concept taught: Spec-Driven Development — "Vague requirements produce flawed results faster; clear specs produce excellent results quickly." The single most important lesson, taught through play.

Scene 4 — Press GO (Work Animation)

Player action: Tap the giant **[GO]** button.

The Support Bot springs to life — reads Mia's message, checks the order, looks up the policy, and types a reply with a gear-filled progress bar.

Support Bot to Mia: "So sorry about the zipper, Mia! I've approved your return and a refund is on the way."

Concept taught: The agent *composes* its Brain + Hands + Training to deliver an outcome — the full Digital FTE in action.

Scene 5 — Verify (Human-in-the-Loop)

A panel shows the Bot's reply with two buttons: **[Approve]** and **[Improve]**.

Nova: "One golden rule, boss: the agent does the work, but YOU verify the outcome. Does this answer look good?"

Player action: Tap **Approve.** Mia sends a heart and a 5-star rating; coins rain down.

Concept taught: The 10-80-10 rule — "Humans define intent. Agents execute. Humans verify outcomes."

Scene 6 — Rewards & The Idle Hook

Reward banner: **+50 Coins**, **+100 XP (Level 1 Complete!)**, **Unlocked: Night Shift** (your bot earns coins while you are away).

Nova: "Amazing! Your first Digital FTE is officially on the job — even after you log off. Ready to grow your factory?"

Concept taught: Autonomous workers create value continuously (the idle-earning loop and retention hook).

Scene 7 — Optional "Learn the Real Thing" Card

A dismissible card: "What you just did has a real name: building a **Digital FTE**. Want the real-world version? See AgentFactory Course 15: 'From Agent to Digital FTE.'"

Concept taught: Bridges the game to the actual AgentFactory crash courses — turning fun into a real learning funnel.

Level 1 Concept Map (Summary)

Scene	Player Action	Real Concept Learned
0	Tap to start	Workforce = business value
1	Build a worker	Mode 1 vs Mode 2
2	Drag 4 parts	Digital FTE anatomy: Brain, Hands, Training, Shift
3	Fill the Spec Card	Spec-Driven Development
4	Press GO	Agent composes capabilities
5	Approve result	10-80-10 rule (verify)
6	Collect rewards	Autonomous 24/7 value
7	Optional link	Bridge to real courses

Part 3 — Full 26-Level Progression Map

The 26 levels are organized into 4 game "Worlds" mirroring AgentFactory's three curriculum tiers (Foundations, General Agents, Manufacturing) plus a Production/Scale endgame.

Note on accuracy: All 26 course titles below are the exact, verified titles from the AgentFactory "Getting Started" crash-course list. The game level names and mechanics are an original design that maps each course to a playable level.

World 1 — "The Startup Garage" (Foundations · Courses 1–6)

Lvl	Game Level Name	Mirrors Course	What You Do	Concept Learned	Unlock
1	Your First Hire	1. What AI Actually Is	Build your first Support Bot (4 parts)	What an AI agent is	Night Shift (idle coins)
2	The Magic Words	2. AI Prompting in 2026	Tune a prompt-dial for better replies	Prompting = clear instructions	Prompt Tuner
3	Markdown Builder	3. Markdown In, HTML Out	Snap markdown blocks into a webpage	Structured input to clean output	Page Builder
4	Code You Never Write	4. Code You Never Write	Describe what you want; agent writes it	AI as your coder	Auto-Builder bench
5	Plug It In	5. Skills & Connectors	Connect tool-chips & skill-books	Skills & Connectors (MCP basics)	Connector slots
6	Think Like a Boss	6. How to Think in the AI Era	Choose intent cards; agent executes	Human sets intent, AI executes	World 2 unlocked

World 2 — "The Open Floor" (General Agents · Courses 7–9)

Lvl	Game Level Name	Mirrors Course	What You Do	Concept Learned	Unlock
7	Code Companion	7. Agentic Coding Crash Course: Claude Code and OpenCode	Pair with a coding agent to fix a bug	Agentic coding	Coder Bot
8	The Knowledge Worker	8. Cowork Crash Course	Deploy a Cowork bot to draft docs & emails	Knowledge-work agents	Cowork desk
9	The Perfect Spec	9. Spec-Driven	Master the	Spec-Driven	MODE

Lvl	Game Level Name	Mirrors Course	What You Do	Concept Learned	Unlock
		Development	Spec Card (specify-clarify-plan-build)	Development	CHOICE

Mode Fork after Level 9: the player picks a playstyle (replayable to do both): Mode 1 ("use AI to do your work" side-levels) or Mode 2 (the Manufacturing campaign, Worlds 3–4, the main game).

World 3 — "The Factory Floor" (Manufacturing · Courses 10–20)

Lvl	Game Level Name	Mirrors Course	What You Do	Concept Learned	Unlock
10	The Problem Solver	10. Problem Solving with General Agents	Send a general agent to crack a tricky customer problem	Problem-solving with general agents (Seven Principles)	Solver Bot
11	The Claw	11. OpenClaw with General Agents	Use OpenClaw to let an agent grab & control on-screen tasks	OpenClaw automation	OpenClaw arm
12	Snake Charmer	12. Python in the AI Era	Drop a Python engine into a worker to crunch data	Python in the AI era	Python core
13	The Searchable Brain	13. Give Your AI Searchable Context: RAG on Postgres with pgvector	Build a searchable knowledge vault the agent queries	RAG (searchable context) with pgvector	RAG vault
14	Build-A-Bot	14. Build AI Agents Crash Course	Assemble a full custom agent	Building agents end-to-end	Custom Agent kit
15	From Worker to Employee	15. From Agent to Digital FTE	Upgrade an agent into a 24/7 Digital FTE	The Digital FTE	FTE badge

Lvl	Game Level Name	Mirrors Course	What You Do	Concept Learned	Unlock
16	The Nervous System	16. From Digital FTE to Production Worker with a Nervous System	Wire workers to an event nervous system (Inngest)	Production worker; events, durability, flow	Event bus
17	Mission Control	17. Building a Workforce with Paperclip	Hire/assign/govern via the Paperclip panel	Workforce management layer	Paperclip console
18	Hiring on Demand	18. From Fixed to Dynamic Workforce	Auto-spawn workers when demand spikes	Runtime hiring under policy	Auto-hire
19	The Delegate	19. From Founder Bottleneck to Owner Delegate	Give yourself a personal delegate agent	Each human gets a delegate	Delegate avatar
20	Quality Inspector	20. Eval-Driven Development for AI Employees	Run agents through a testing gauntlet	Eval-Driven Development	Eval lab

World 4 — “The Empire” (Production & Scale · Courses 21–26)

Lvl	Game Level Name	Mirrors Course	What You Do	Concept Learned	Unlock
21	To the Cloud	21. Deploy Your Agent Harness to the Cloud	Ship your factory to the cloud (Docker/K8s/Dapr/Ray)	Cloud-native deployment	Cloud city
22	The Architect	22. Choosing Agentic Architectures	Pick the right blueprint/architecture for each worker	Choosing agentic architectures	Architecture board
23	Show Me the Money	23. Payment-Enabled Agents: ACP, AP2, x402, and MPP	Let agents handle real payments safely	Payment-enabled agents	Payments rail
24	The Recruiter	24. Which AI Employees Should You Use	Pick the best AI employee for each role	Selecting AI employees	Roster screen

Lvl	Game Level Name	Mirrors Course in 2026?	What You Do	Concept Learned	Unlock
25	Cheat Codes	25. Cheatsheets	Unlock quick-reference power-up cards	Cheatsheets / quick reference	Cheatsheet deck
26	Master Engineer	26. Agentic Engineering Fundamentals	Run a full autonomous company with all principles	Capstone: Agentic Engineering Fundamentals	Endgame + Prestige

Progression Design Notes

- **Difficulty curve:** Worlds 1–2 are tutorial-easy (can't lose). World 3 adds light strategy (managing multiple workers). World 4 adds optimization/endgame depth.
- **Pacing:** Levels 1–9 ~3 min each. Levels 10–26 grow to 5–8 min with the factory-management layer.
- **Replay value:** the Mode 1 / Mode 2 fork plus Prestige at Level 26 (restart with bonuses) keep players returning.
- **Learning funnel:** every milestone level gently links to the matching real AgentFactory course, turning the game into an on-ramp for the actual curriculum.

Part 4 — Design Specification: Fun, Easy, Interactive, Beautiful

This section is not adjectives — it is a buildable spec. Each of the four pillars is grounded in an established, named design framework and turned into concrete decisions a developer can implement on day one.

Pillar 1 — FUN (grounded in Lazzaro's "4 Keys to Fun" + Flow + Game Feel)

Nicole Lazzaro's research identifies four distinct sources of fun. A game that only hits one feels thin; *Agent Factory Tycoon* deliberately delivers all four.

Key to Fun	What it is	How the game delivers it
Easy Fun	Curiosity, novelty, play	New worker types, tools, and worlds drip-fed every level; free-explore sandbox mode
Hard Fun	Challenge, mastery, "fiero"	Daily challenges, eval-scoring (Lvl 20), Prestige loop, optimizing a factory for max output

Key to Fun	What it is	How the game delivers it
	(triumph)	
Serious Fun	Play with real-world meaning	Every mechanic teaches a real AgentFactory concept the player can actually use
People Fun	Social, sharing, competition	Leaderboards, shareable factory layouts, “beat my Support FTE” challenges

Flow: difficulty must track skill (Csikszentmihalyi)

Fun collapses when a game is too hard (anxiety) or too easy (boredom). We hold the player in the **flow channel** with:

- **Dynamic Difficulty Adjustment (DDA):** if a player retries a level 3+ times, Nova offers a hint and the next customer request gets simpler. If they breeze through, optional “expert orders” appear for bonus coins.
- **A clean ramp:** Worlds 1–2 are unloseable tutorials; challenge is introduced only after the core loop is automatic.

Game Feel (“juice”): the single biggest multiplier on perceived fun

Every player action gets disproportionate, satisfying audiovisual feedback. This is the **mandatory juice checklist** for the MVP:

- Snap-to-slot with a spring overshoot (DOTween ease-out-back) + “click” SFX + a 6-particle sparkle burst
- Button presses scale to 0.92 then bounce back; coins fly to the counter on an arc and the counter rolls up
- Confetti + screen-flash on level-up; gentle screen-shake (max 4px) on big wins only
- The agent mascot has idle micro-animations (blink, breathe, look around) so the screen is never “dead”
- The “silly spec” gag (Level 1) — failure is *funny*, never punishing — is the emotional signature of the game

Pillar 2 — EASY (grounded in Cognitive Load Theory, “Show-Don’t-Tell,” Fitts’s Law, Krug)

One new mechanic per level (Cognitive Load Theory)

The 26-level map already enforces this: each level introduces exactly **one** new concept. We never stack two new mechanics in the same level. This is the core reason the game stays easy as it deepens.

Teach by doing, not by reading (“Show, Don’t Tell”)

Modeled on Nintendo’s level-design philosophy (e.g., the wordless teaching of World 1-1 in Super Mario Bros.):

- **No text tutorials.** Nova speaks one sentence, then the *only* tappable thing on screen is the action we want to teach. The player learns by succeeding.
- **Just-in-time hints:** guidance appears at the moment of need, then disappears. Nothing is front-loaded.
- **Diegetic teaching:** the “silly spec” outcome teaches the spec lesson through consequence, not a tooltip.

Forgiveness (remove anxiety)

- **No hard fail states, no timers in Worlds 1–2, no lost progress.** The worst outcome is “your agent could do better — tweak and retry.”
- Auto-save after every loop; the player can never lose a worker they built.

Reachability (Fitts’s Law + Krug’s “Don’t Make Me Think”)

- **Mobile-first thumb zones:** all primary controls sit in the bottom 40% of the screen; the GO button is the largest target on screen.
- **Minimum 48x48dp tap targets**, generous spacing, no precision dragging required (slots have magnetic snap).
- **Self-evident UI:** icons + one-word labels; if an element needs a paragraph to explain, it is redesigned.

Accessibility (built in, not bolted on)

- Colorblind-safe palette (never color-only signaling — pair color with icon/shape)
- Adjustable text size; high-contrast mode
- One-handed portrait play; full captions for all spoken Nova lines
- Reduced-motion toggle (disables screen-shake/heavy particles)

Pillar 3 — INTERACTIVE (grounded in Direct Manipulation + Player Agency)

Direct manipulation, not menus

Players *touch the world*, not abstract menus. Every core verb is a physical gesture:

Player intent	Gesture	Immediate response
Build a worker	Drag part to slot	Magnetic snap + sparkle + part animates on
Give a job	Tap word-tiles onto a card	Card fills, glows green when valid

Player intent	Gesture	Immediate response
Run the worker	Tap GO	Live work animation plays out
Judge the result	Tap Approve / Improve	NPC reacts (rating, emoji), economy updates
Grow the factory	Drag-place desks/rooms	Building snaps to grid, workers move in

Meaningful choices = agency

- **The Mode Fork (Level 9):** a genuine branch (Mode 1 vs Mode 2) that changes the next stretch of play and is replayable.
- **Build choices matter:** which Brain/tools/skills you pick changes how well the worker performs on a given order — there's a "right-ish" build to discover, not one forced answer.

A living, reactive world

- NPC customers queue, tap their feet, and react in real time to good/bad service.
- Deployed workers visibly bustle around the factory floor between orders (the idle layer is *visible*, not just a number).

Pillar 4 — GOOD GRAPHICS (a concrete, achievable, mobile-friendly art direction)

These are definitive choices, not options — chosen for charm, readability, and the ability to ship on web + mobile at 60fps.

Chosen style: stylized **2.5D low-poly isometric**, rendered in 3D

Justification: low-poly reads beautifully at any resolution, is cheap to render on phones, animates with personality, and has mature ready-made asset libraries — so a small team can hit "good graphics" without a large art budget. Reference charm: *Two Point Hospital*, *Mini Motorways*, *Monument Valley*.

Definitive color palette (tied to the AgentFactory brand)

Role	Hex	Use
Primary Violet	#6D28D9	Brand, headers, the "Brain" orb
Teal Accent	#14B8A6	Interactive highlights, the "Hands" tools
Coin Gold	#FBBF24	Rewards, currency, celebration
Success Green	#22C55E	Valid specs, approvals
Playful Red	#EF4444	The funny "wrong" feedback (never harsh)
Cream Base	#FFF7ED	Backgrounds, calm canvas
Ink	#1E293B	Text, outlines

Rule: warm tones (gold/green) signal reward; cool tones (violet/teal) signal interactivity; red is reserved for *playful* error only.

Production pipeline (real, off-the-shelf)

- **Engine:** Unity 6 LTS with the Universal Render Pipeline (URP) — best balance of mobile performance + WebGL export to match AgentFactory's no-install labs.
- **Art assets:** Synty POLYGON / Kenney.nl low-poly packs as a base kit (fast, cohesive, low-cost), customized to brand.
- **Animation juice:** DOTween for UI/tween motion; skeletal mascot animation following Disney's 12 principles — lean on squash-and-stretch, anticipation, and follow-through for life.
- **Lighting & post:** baked global illumination + a soft directional key light; post-processing stack of gentle Bloom, slight Vignette, and Color Grading for a warm, polished look.
- **UI:** soft-shadow, rounded "claymorphic" components; TextMeshPro for crisp text at every resolution.

Performance targets (so "good graphics" actually ships)

- 60fps on a mid-range 3–4-year-old Android phone
- Initial WebGL load under ~30 MB; assets streamed thereafter
- Aggressive static/dynamic batching and texture atlasing to keep draw calls low
- Reduced-motion and "lite graphics" toggles for low-end devices

How We'll Know It Works (validation, not vibes)

Pillar	Measured by	Target
Easy	Tutorial (Level 1) completion rate	> 90% finish without quitting
Fun	Day-1 retention / sessions per day	Players return same day; multiple short sessions
Interactive	Avg. actions per minute in the build loop	High touch frequency, low idle-confusion time
Graphics	Frame rate on target device + qualitative "delight" playtests	Stable 60fps; testers smile at the juice
Learning	Can a player, after World 1, correctly name the 4 FTE parts?	> 80% recall in a quick post-play quiz

If Level 1 hits >90% completion and testers smile at the assemble-spec-verify loop, the concept is validated — then scale to all 26 levels.

Part 5 — Storyboard: Scene 2, The Worker Builder

Screen: Portrait mobile · **Goal:** drag 4 parts into the worker · **Feel:** tactile, magnetic, celebratory

This is the core mechanic screen, drawn as a 6-frame sequence. Each frame lists the visual, the juice/animation, the audio, Nova's single line, and the concept it teaches.

Persistent Screen Layout (the "stage")

```
+-------------------------------------+
| < Back        LEVEL 1      o . . . | header: 4-dot part-progress tracker
+-------------------------------------+
|            A S S E M B L Y           |
|                                      |
|              +--------+              |
|              : ? ?   :   <- BRAIN    | upper area =
|              +--------+      slot    | the worker
|           +------------+             | (slots glow
|      (?)--|   B O D Y   |--(?)       | dashed when
|       ^   :  +------+   :   ^        | empty)
|     HANDS : : ? ? :     :  HANDS     |
|           : +-----+     :            |
|           +---[ ? ? ]--+ <- TRAINING|
|              /        \              |
|             [  ? ?  ] <- SHIFT       |
|                                      |
+- - - - - - Nova hint - - - - - - - - +
|  (o) "Drag the Brain into your       |
|       worker to give it a mind!"     |
+============ THUMB ZONE =============+
|   +-----+  +-----+  +-----+  +-----+   | draggable
|   | Br |  | Hn |  | Tr |  | Sh |   | part trays
|   +-----+  +-----+  +-----+  +-----+   | (bottom 40%,
|   Brain   Hands  Train.  Shift      | thumb-reachable)
+-------------------------------------+
```

Frame 1 — The Bench Opens

```
Header:  o . . .   (0 of 4 parts placed)
Worker:  empty silhouette, all 4 slots pulsing with a soft dashed glow
Nova:   "Every worker needs 4 parts. Let's start with the Brain!"
Trays:   [Brain] tray gently bounces up/down to invite the first drag;
         the other 3 trays are dimmed (greyed ~40%)
```

- **Juice:** the empty worker does a slow "breathe" scale (1.0 to 1.02). Only the Brain tray is lit, so attention has exactly one target.

- **Audio:** soft ambient hum plus a single gentle chime as the screen settles.
- **Concept:** there is a *structure* to an AI worker — four parts, introduced one at a time (one-new-thing rule).
- **Accessibility:** the active tray also shows an arrow plus label, not color alone.

Frame 2 — Dragging the Brain (finger down)

```
              +--------+
              : (o)(o):  <- BRAIN slot now glows BRIGHT
              +--------+      (it is the live drop target)
                  ^
                [Brain] <- orb lifts off the tray, enlarges 1.15x,
                           trails sparkles, and follows the finger
Nova: "Drop it on the head!"
```

- **Juice:** on pick-up the orb pops (scale 1.0 to 1.15, ease-out-back) and casts a soft drop-shadow so it feels lifted. A faint dotted line connects orb to target slot. The matching slot brightens; non-matching slots stay dim, so the player cannot feel lost.
- **Audio:** a rising "whoop" pitch while held (tracks height) — playful, like lifting a magnet.
- **Concept:** the **Brain = the reasoning model** (Claude/GPT/Gemini). Head placement makes "thinking" literal.

Frame 3 — SNAP! Brain Locks In

```
Header:  o o . .   (1 of 4)
              +--------+
              : (o_o) :  <- eyes blink ON; the worker now has a face
              +--------+
              *   *   *   <- 6-particle sparkle burst
Nova: "Nice! A mind online. Now give it Hands."
Trays:   [Brain] consumed/checked · [Hands] now lights up and bounces
```

- **Juice (the signature moment):** magnetic snap with a spring overshoot (DOTween ease-out-back), 6 sparkle particles, a tiny 3px screen-pulse, and the worker's eyes blink open — instantly giving it personality. The progress dot fills with a satisfying tick.
- **Audio:** a crisp "click-ding" — the core reward sound the whole game is built around.
- **Concept:** placement leads to immediate, visible consequence. The worker visibly became smarter.

Frame 4 — Hands Attach

```
Header:  o o o .   (2 of 4)
           +------------+
   [Ord][Ch]|   B O D Y  |[Ch][Ord]  <- tool-arms unfold
           :  (o_o)     :              (Orders + Chat)
```

```
      +-----------+
Nova: "Hands let it DO things — touch real systems."
```

- **Interaction:** the Hands tray holds multiple tool-chips (Orders, Chat). The player drags the two the job needs — a first taste of meaningful build choice. Each snaps with the same click-ding plus sparkle.
- **Juice:** little articulated arms unfold with squash-and-stretch and a mechanical "whirr."
- **Concept: Hands = MCP + tools** that let the agent act on real systems.

Frame 5 — Training Book Absorbed

```
Header:  o o o .   (3 of 4, last dot about to fill)
           :  +-----+  :
           :  : Book :  :  <- book flips pages, then dissolves
           :  +-----+  :     into the worker (knowledge "learned")
Nova: "Training = what it KNOWS. Here's your Returns Policy."
```

- **Juice:** the skill-book opens, pages riffle (a quick flip animation), then it glows and dissolves *into* the worker — a visual metaphor for learning. The worker gives a small approving nod.
- **Audio:** a soft page-riffle plus a warm "absorb" shimmer.
- **Concept: Training = Agent Skills** (domain knowledge plus SOPs).

Frame 6 — The Shift Clock: Worker Complete

```
Header:  o o o o   (4 of 4 — full!)
              [ 24/7 clock ] <- snaps to the base, hands spin once
              (o_o) party    <- worker powers up fully, waves
            *  * CONFETTI *  *
Nova: "Your first Digital FTE is ALIVE! Now give it a job."
                       [  CONTINUE  >  ]  <- appears, pulsing
```

- **Juice:** the final snap triggers a full celebration — confetti rain, a one-time gentle screen-shake (<=4px), the worker stands up, eyes go happy, and it gives a little wave. All four progress dots do a victory shimmer.
- **Audio:** a triumphant rising "power-up" arpeggio — the triumph/"fiero" beat (Hard Fun payoff).
- **Transition:** a single glowing **CONTINUE** button slides into the thumb zone — the only tappable element — leading into Scene 3 (the Spec Card).
- **Concept: Shift = autonomous 24/7 operation.** The worker is now a complete Digital FTE: Brain + Hands + Training + Shift.

Designer Notes for This Screen

- **One target at a time:** only the next valid slot/tray is ever lit; everything else dims. This is how the screen stays easy despite four steps — the player is never hunting.

- **Magnetic, forgiving drops:** slots have a wide snap radius; near-misses still lock in. No precision required (Fitts's Law).
- **Wrong-part feedback is gentle:** dragging the Shift clock onto the head makes it bounce off with a soft "boing" and a smiling shake — no penalty.
- **Every snap = the same reward chord** (click-ding plus sparkle plus dot fill). Consistency builds the satisfying rhythm.
- **The face is the hook:** the worker gains eyes at the Brain step and emotion by the end — players bond with a character, not a form.
- **Reduced-motion mode:** confetti/shake replaced by a soft glow pulse; all timings and SFX preserved.

Part 6 — Storyboard: Scene 3, The Spec Card (with Nova)

Screen: Portrait mobile · **Goal:** fill a fill-in-the-blank job order with word-tiles · **Teaches:** Spec-Driven Development · **Signature beat:** the "silly spec" gag

Same format as the Scene 2 storyboard: a 6-frame sequence, each beat noting visual, juice, audio, Nova's line, and the concept.

Persistent Screen Layout (the "stage")

```
+-------------------------------------+
| < Back      LEVEL 1     SPEC CARD   |
+-------------------------------------+
|   Your worker stands ready, looking |
|   at you, waiting for orders        |
|             (o_o)?                   |
|                                     |
|   == JOB ORDER ==================   |
|   |  My worker should           |   |
|   |     +--------------+        |   | <- blank #1
|   |     |   _______    |        |   |
|   |     +--------------+        |   |
|   |  when                       |   |
|   |     +--------------+        |   | <- blank #2
|   |     |   _______    |        |   |
|   |     +--------------+        |   |
|   |  then                       |   |
|   |     +--------------+        |   | <- blank #3
|   |     |   _______    |        |   |
|   |     +--------------+        |   |
|   ==============================    |
+- - - - - - Nova hint - - - - - - - +
|  (o) "Tell your worker its JOB.     |
|       Tap a word to drop it in!"    |
+=========== WORD BANK ==============+
```

```
|  [help the customer] [a return      |
|   request arrives] [confirm the     | <- tappable
|   refund]  [panic]  [eat lunch]     |    word-tiles
|  [ignore everyone]                  |
+-------------------------------------+
```

Frame 1 — The Card Slides Up

```
Worker:  stands center-stage, tilts head, a "?" bubble floats above
Card:    JOB ORDER slides up from the bottom with a paper "whoosh";
         all 3 blanks pulse with a soft dashed glow
Nova:    "Tell your worker its JOB. Tap a word to drop it in!"
Bank:    6 word-tiles fan out like cards being dealt
```

- **Juice:** the card has a subtle paper texture and a gentle float/bob. Word-tiles deal in with a riffle (staggered 40ms each). Blank #1 glows brightest — the suggested starting point.
- **Audio:** paper "whoosh" plus a soft card-deal riffle.
- **Concept:** a worker needs *explicit instructions* — you are about to write its spec.
- **Accessibility:** blanks are numbered 1-2-3 and labeled, not just color-glowing.

Frame 2 — Tapping a Word-Tile

```
Bank:   player taps [panic] ->
        tile lifts 1.1x, glows, and arcs up toward blank #1
        +-------------+
        |   panic >   |  <- flies into the slot
        +-------------+
Nova:   "Hmm... bold choice."
```

- **Interaction:** tap-to-fill (one-tap, no precise dragging — easy). The tile flies on an arc into the next open blank and snaps with a soft click. Tap a filled blank to pop the word back out and try again.
- **Juice:** arc tween (ease-out-back), tile snaps with a 4-particle puff. Nova's eyes follow the tile.
- **Audio:** light "pop-click" as it lands.
- **Concept:** specs are built piece by piece — and you can revise freely (no penalty for experimenting).

Frame 3 — The "Silly Spec" (the signature gag)

```
Card filled (nonsense):
   My worker should  [ panic ]
   when              [ eat lunch ]
   then              [ ignore everyone ]
   -> card flashes amber, a [ TRY IT > ] button appears
Player taps TRY IT...
```

```
Worker reaction:
        (x_x)  <- eyes go spiral
       \(o_o)/   spins in circles, papers fly everywhere,
        _|_|_     trips over its own tool-arms
   *sad trombone: womp-womp-wommmp*
Nova (laughing): "HA! See? Fuzzy orders = total chaos.
                  Let's give it a REAL spec."
```

- **Juice:** full slapstick — the worker spins, juggles papers, a tiny rain-cloud appears over its head. Exaggerated squash-and-stretch. This is failure made delightful — never punishing.
- **Audio:** the famous sad-trombone "womp-womp," comedic boings, scattered paper rustle.
- **No penalty:** no coins lost, no life lost. The card simply shakes its head and resets the blanks, tiles flying back to the bank.
- **Concept (the core lesson, taught by consequence):** *vague/wrong specs produce chaos faster.* The player feels why clarity matters instead of being told — this is the heart of Spec-Driven Development.

Frame 4 — Nova Coaches the Retry

```
Card:   blanks empty again, gently pulsing
Nova:   "A good spec answers: DO what? WHEN? then WHAT?
         The right words are glowing — give it a go!"
Bank:   the three CORRECT tiles now have a faint helpful shimmer
        [help the customer] [a return request arrives]
        [confirm the refund]   <- these subtly glow
```

- **Just-in-time help (DDA):** only on a *failed* attempt does Nova nudge by softly highlighting the sensible tiles. A first-try-correct player never sees the hint — preserving challenge for the capable.
- **Juice:** the helpful tiles do a slow breathing glow; the silly tiles fade slightly back.
- **Concept:** a clear spec has a shape — **action then trigger then outcome.**

Frame 5 — The Perfect Spec

```
Card filled (clear):
   My worker should  [ help the customer ]
   when              [ a return request arrives ]
   then              [ confirm the refund ]
   -> every blank locks GREEN, card emits a warm glow
        =============================
        |  (check)  SPEC LOCKED IN!  |  <- green seal stamps on
        =============================
Worker:  nods confidently (o_o)v — "got it, boss!"
Nova:    "Crystal clear! THAT'S a spec a worker can nail."
```

- **Juice:** as the last correct tile lands, the whole card sweeps green left-to-right, a "SPEC LOCKED IN" wax-seal stamps down with a satisfying thump, and the worker straightens up and salutes. Coins shimmer in anticipation.
- **Audio:** a bright ascending "success" chord plus a stamp "ka-chunk."
- **Concept: clear specifications produce excellent results** — the validated, well-formed spec (action/trigger/outcome).

Frame 6 — Hand-off to "GO"

```
Card:   slides up and "attaches" to the worker (it now carries its job)
A single glowing button slides into the thumb zone:
        +--------------------------+
        |    >   SEND IT TO WORK    |  <- only tappable element
        +--------------------------+
Nova:   "Spec's ready. Hit GO and watch your worker shine!"
```

- **Transition:** the locked card folds into the worker's "Training" area, visually linking spec to worker. The lone **SEND IT TO WORK** button leads into Scene 4 (Press GO / the work animation).
- **Audio:** a soft "ready" chime; button pulses invitingly.
- **Concept:** the spec now *drives* the worker — intent set, ready to execute (the start of the 10-80-10 loop).

Designer Notes for This Screen

- **Tap, don't drag:** one-tap-to-fill keeps it effortless on mobile; tap a filled blank to undo. Zero precision required.
- **Failure is the best teacher here:** the silly-spec gag is the emotional and educational centerpiece — make the slapstick generous and the reset instant and friendly.
- **Hints are earned, not forced:** correct-tile glow appears only after a wrong attempt (Dynamic Difficulty) — capable players keep the challenge.
- **Shape over vocabulary:** the card visibly teaches the action-trigger-outcome pattern, which transfers directly to real Spec-Driven Development.
- **Consistent reward chord:** the green "lock-in" reuses the game's signature success sound, tying it to every other win.
- **Reduced-motion mode:** the slapstick spin becomes a gentle wobble plus cloud icon; sad-trombone and all SFX preserved.

Part 7 — Storyboards: Scene 4 (Press GO) and Scene 5 (Verify)

These two scenes complete the Level 1 visual sequence. Scene 4 is the “80%” of the 10-80-10 loop — the agent executes. Scene 5 is the final “10%” — the human verifies. Same format as the earlier storyboards.

Scene 4 — Press GO and the Live Work Animation

Screen: Portrait mobile · **Goal:** watch the worker execute the spec · **Teaches:** an agent composes Brain + Hands + Training to deliver an outcome

Persistent Screen Layout (the “stage”)

```
+------------------------------------+
| < Back      LEVEL 1     AT WORK     |
+------------------------------------+
|   Chat with Mia                     |
|   +------------------------------+  |
|   | Mia: my backpack zipper is   |  |
|   |      broken, can I return?   |  |
|   +------------------------------+  |
|                                     |
|        (o_o)   <- worker at desk    |
|       /[==]\      typing...         |
|        _||_                         |
|                                     |
|   [#### Working ... 60% ####----]   |
|   read -> check order ->            |
|   look up policy -> write reply     |
+- - - - - - Nova hint - - - - - - - +
|  (o) "Watch your worker go. You set |
|      it up great!"                  |
+------------------------------------+
```

Frame 1 — Press GO

```
The SEND IT TO WORK button (from Scene 3) sits in the thumb zone, pulsing.
Player taps it:
   button squashes to 0.92x, then bounces back, emits a ring of sparks
   the worker snaps upright, cracks its knuckles (tool-arms flex)
Nova:  "Here we go!"
```

- **Juice:** button press with spring-back; a quick “power-on” sweep washes across the worker; the screen subtly zooms in on the worker’s desk.
- **Audio:** a satisfying “ka-chink” press plus a rising “engine start” whir.
- **Concept:** the human sets intent and presses go — the agent now takes over (start of the 80%).

Frame 2 — The Worker Reads (Brain + Chat)

```
Mia's message highlights with a soft scan-line sweep
+-------------------------------+
| Mia: ...broken... return? ...|  <- words light up as "read"
+-------------------------------+
     (o_o)  <- worker's eyes flick left-to-right (reading)
Step ticker: [ READING THE REQUEST ]
```

- **Juice:** a scan-line passes over Mia's text; the worker's eyes track it. A small thought-bubble shows a tiny brain icon pulsing.
- **Audio:** soft "blip-blip" scanning tones.
- **Concept:** the **Brain** comprehends the request via the **Chat** tool (Hands).

Frame 3 — The Worker Acts (Hands + Training)

```
Tool icons light up in sequence as the worker uses them:
   [ Orders ] -> spins, finds Mia's order  (CHECK)
   [ Policy ] -> book glows, rule found     (CHECK)
Step ticker: [ CHECKING ORDER ]  ->  [ LOOKING UP POLICY ]
     (o_o)  <- worker glances at each tool as it fires
```

- **Juice:** each tool-chip lights, does a little spin, then stamps a green CHECK. A connecting line briefly links Brain to each tool — showing the parts working together.
- **Audio:** mechanical "whirr-clunk" per tool, a soft "ding" on each check.
- **Concept: Hands (MCP/tools)** act on real systems; **Training (Skills)** supplies the rule. The Digital FTE composes its parts.

Frame 4 — The Worker Composes the Reply

```
A reply bubble types itself out, word by word:
+-------------------------------+
| "So sorry about the zipper-   |  <- text appears live
|  ..."                         |
+-------------------------------+
[######## Working ... 90% ########--]
     (o_o)  /[==]\  <- hands "type" on a little keyboard
```

- **Juice:** live typewriter text; the progress bar fills with rolling gear icons; tiny key-tap particles fly off the worker's hands.
- **Audio:** gentle mechanical-keyboard clicks; the progress bar emits a soft rising hum as it nears 100%.
- **Concept:** the agent synthesizes everything it gathered into a single, grounded outcome.

Frame 5 — Reply Sent, Hand-off to Verify

```
Progress hits 100% -> a soft "ping"
The finished reply slides into the chat as a draft (not yet sent to Mia)
     (o_o)v  <- worker leans back, done, looks to you
```

```
   A panel begins sliding up: "Review your worker's reply"
Nova:  "Nice work by your FTE! One last step - let's check it."
```

- **Juice:** progress bar flashes full-green; the worker exhales (small relief animation) and turns to face the player, handing off control.
- **Audio:** a completion "ping" and a soft swoosh as the Verify panel rises.
- **Concept:** the agent finished the 80% — now control returns to the human for the final 10%.

Scene 5 — Verify (Approve or Improve)

Screen: Portrait mobile · **Goal:** approve a good result (or send it back to improve) · **Teaches:** the 10-80-10 rule — humans verify outcomes

Persistent Screen Layout (the "stage")

```
+-----------------------------------+
| < Back      LEVEL 1     VERIFY      |
+-----------------------------------+
|   Your worker drafted a reply:      |
|   +-----------------------------+   |
|   | "So sorry about the zipper,  |   |
|   |  Mia! I've approved your     |   |
|   |  return and a refund is on   |   |
|   |  the way."                   |   |
|   +-----------------------------+   |
|                                     |
|   Does this look good, boss?        |
+- - - - - - Nova hint - - - - - - - +
|  (o) "The agent does the work, but  |
|       YOU verify. Golden rule!"     |
+========== THUMB ZONE ==============+
|   [   IMPROVE   ]   [   APPROVE   ] |
+-----------------------------------+
```

Frame 1 — The Verify Panel Slides Up

```
   The draft reply card settles center-screen with a gentle bob.
   Two buttons rise into the thumb zone:
      [ IMPROVE ] (amber, left)     [ APPROVE ] (green, right)
        (o_o)  <- worker waits beside the card, hopeful
Nova:  "The agent did the work - now YOU verify the outcome."
```

- **Juice:** the reply card has a subtle highlight outline inviting a read; the two buttons pulse alternately so neither is missed; the worker watches with hopeful eyes.
- **Audio:** soft panel swoosh; a quiet expectant tone.
- **Concept:** verification is a deliberate, named step — not an afterthought.

Frame 2 — Nova Frames the Golden Rule

```
Nova (one beat, then steps aside):
   "Humans set intent. Agents execute. Humans verify.
    That's the rhythm of every great AI team."
   The two buttons remain; nothing else is tappable.
```

- **Just-in-time teaching:** Nova states the 10-80-10 rule once, exactly when the player is about to do it, then gets out of the way.
- **Concept: 10-80-10** — the core operating loop of working with Digital FTEs.

Frame 3 — The IMPROVE Branch (optional)

```
Player taps [ IMPROVE ]:
   a quick coaching popup: "What should it do better?"
   tap a fix-chip: [ warmer tone ] [ add refund ETA ] [ shorter ]
        (o_o)? -> (o_o) -> worker re-drafts in ~2s
   the reply updates; player returns to the Approve choice
Nova:  "Good eye! Small tweaks make a big difference."
```

- **Juice:** the worker nods, quickly re-types (a fast version of the Scene 4 typing), and the card refreshes with a soft glow.
- **No penalty:** improving is rewarded, not punished — it models healthy human-in-the-loop iteration.
- **Concept:** verification can mean *steering*, not just yes/no — the human refines the outcome.

Frame 4 — The APPROVE Branch

```
Player taps [ APPROVE ]:
   green button stamps a big CHECK; the draft "sends" to Mia (swoosh)
   +--------------------------------+
   | Mia: Thank you!! <3  *****     |  <- heart + 5-star rating
   +--------------------------------+
        (o_o)  ->  (o_^)  <- worker beams, takes a little bow
   Coins arc from Mia's message to the counter: +50 (cha-ching!)
Nova:  "Boom! Happy customer, happy boss. That's the loop!"
```

- **Juice:** approve-stamp thump; Mia's heart pops and 5 stars fill one by one with rising chimes; coins fly on an arc and the counter rolls up; a small confetti puff.
- **Audio:** the signature "click-ding" success chord, star-fill arpeggio, coin "cha-ching."
- **Concept:** a verified, approved outcome closes the loop — value delivered.

Frame 5 — Hand-off to Rewards

```
   The screen begins its transition to Scene 6 (Rewards):
   a banner peeks up - "+50 Coins ... +100 XP ..."
        (o_^)  <- worker waves, heads to its desk to keep working
Nova:  "Your first Digital FTE is officially on the job. Let's tally it up!"
```

- **Transition:** smooth wipe into the reward banner (Scene 6), with the worker walking off to the factory floor to begin its autonomous shift.
- **Concept:** one full loop complete (intent -> execute -> verify -> reward); the worker now runs on its own.

Designer Notes for These Two Screens

- **Show the parts working together (Scene 4):** lighting up Brain -> Hands -> Training in sequence is the payoff for the assembly in Scene 2 — players literally see what they built come alive.
- **Never a dead screen:** the worker always has micro-animation (reading, typing, waiting, beaming) so the execute phase feels alive, not like a loading spinner.
- **Verify is a real choice (Scene 5):** Approve and Improve are both first-class, both rewarded — this teaches that verification is active, not a rubber stamp.
- **Consistent reward grammar:** the same click-ding, coin arc, and star chimes used everywhere tie Scene 5's payoff to every other win in the game.
- **Reduced-motion mode:** typing/confetti become soft fades; all audio cues and timings preserved.

Part 8 — Storyboards: Scene 6 (Rewards) and Scene 7 (Learn the Real Thing)

These two scenes close out Level 1. Scene 6 is the celebration and progression payoff (coins, XP, the Night Shift unlock). Scene 7 is the optional bridge that connects the play to the real AgentFactory curriculum. Same format as the earlier storyboards.

Scene 6 — Rewards and the Idle Hook

Screen: Portrait mobile · **Goal:** celebrate the win and unlock progression · **Teaches:** autonomous workers create value continuously (the Digital FTE promise)

Persistent Screen Layout (the "stage")

```
+-------------------------------------+
| < Back      LEVEL 1   COMPLETE!      |
+-------------------------------------+
|         * * * LEVEL CLEAR * * *      |
|                                      |
|      +-------------------------+     |
|      |  +50  Coins             |     |
|      |  +100 XP   -> LVL 1 UP |     |
|      |  [#########----] 80%   |     |  <- XP bar filling
|      +-------------------------+     |
|                                      |
```

```
|   UNLOCKED:                             |
|      +-----------------------+          |
|      | (moon) NIGHT SHIFT     |          |
|      | Your FTE earns coins   |          |
|      | while you are away!    |          |
|      +-----------------------+          |
|                                         |
+- - - - - - Nova hint - - - - - - - - +
|  (o) "Your worker keeps earning,       |
|       even after you log off!"         |
+========== THUMB ZONE ==============+
|        [   CONTINUE   >   ]           |
+--------------------------------------+
```

Frame 1 — The Victory Banner

```
   A "LEVEL CLEAR" banner drops from the top with a bounce.
   Confetti rains; the worker (now on the factory floor) waves up at you.
        (o_^)/  <- worker celebrating
Nova:  "You did it, boss - your first Digital FTE is live!"
```

- **Juice:** banner drop with spring-bounce; confetti burst; a brief triumphant zoom on the smiling worker.
- **Audio:** the triumphant "fiero" power-up arpeggio (the big-win chord).
- **Concept:** completion and triumph (Hard Fun payoff) — the player earned this.

Frame 2 — Coins and XP Tally

```
   Rewards count up one by one (not all at once):
      +50 Coins    -> coins arc into the counter, which rolls up
      +100 XP      -> the XP bar fills left-to-right
   [############------] 80%  ->  [#################] 100%  LEVEL UP!
```

- **Juice:** sequential reveal (coins first, then XP) so each lands as its own little hit; the XP bar fills and flashes gold on LEVEL UP; numbers roll rather than snap.
- **Audio:** coin "cha-ching" cascade, a rising tick as the XP bar fills, a "level-up" sting at 100%.
- **Concept:** visible progression — clear evidence of growth keeps players motivated (drip-fed mastery).

Frame 3 — The Night Shift Unlock

```
   A new card flips into view with a shimmer:
      +-----------------------+
      | (moon)  NIGHT SHIFT    |
      | Your FTE now earns     |
      | coins 24/7 - even      |
      | while you are away.    |
      +-----------------------+
```

```
   The worker's desk gains a little clock that starts ticking.
Nova:  "This is the magic of a Digital FTE - it never sleeps."
```

- **Juice:** the unlock card flips in (3D card-flip) with a shimmer sweep; the worker's desk-clock begins a gentle tick; a soft "+coins/hr" tag appears.
- **Audio:** a magical "unlock" chime plus a soft ticking clock.
- **Concept: autonomous 24/7 operation** — the idle-earning hook *is* the Digital FTE promise (and the retention mechanic).

Frame 4 — Hand-off

```
   A single glowing button slides into the thumb zone:
        [   CONTINUE   >   ]
   Tapping it begins the transition toward the next beat
   (the optional "Learn the Real Thing" card, then Level 2).
Nova:  "Ready to grow your factory? Let's keep going!"
```

- **Juice:** the CONTINUE button pulses invitingly; a subtle parallax of the factory floor hints at more to come.
- **Audio:** soft "ready" chime.
- **Concept:** momentum into the next loop — the core gameplay rhythm repeats and scales.

Scene 7 — The "Learn the Real Thing" Bridge Card

Screen: Portrait mobile (a small, dismissible overlay) · **Goal:** connect the fun to the real AgentFactory curriculum · **Teaches:** what the player just did has a real name and a real course

Persistent Screen Layout (the "stage")

```
+-------------------------------------+
|        (the factory floor, dimmed)   |
|                                      |
|   +------ LEARN THE REAL THING --+   |
|   |                              |   |
|   |  What you just built has a   |   |
|   |  real name: a DIGITAL FTE.   |   |
|   |                              |   |
|   |  Want the real-world version?|   |
|   |  -> AgentFactory Course 15:  |   |
|   |     "From Agent to           |   |
|   |      Digital FTE"            |   |
|   |                              |   |
|   |  [ Maybe later ] [ Show me > ]|  |
|   +------------------------------+   |
|                                      |
|  (o) Nova: "Curious how this works   |
```

```
|       for real? Peek anytime."      |
+-----------------------------------+
```

Frame 1 — The Card Gently Appears

```
   The factory floor dims softly; a friendly card slides up
   from the corner (NOT a full-screen takeover - easy to ignore).
        (o_o)  <- Nova peeks in from the side
Nova:  "Psst - want to see the real thing behind this game?"
```

- **Juice:** background dims to ~60%; the card eases in from a corner with a soft shadow; Nova pokes in playfully rather than blocking the screen.
- **Audio:** a gentle, curious "ding" - inviting, never interruptive.
- **Concept:** the game is a doorway, not a wall - the bridge is optional and low-pressure.

Frame 2 — The Connection Is Made

```
   The card names the concept and maps it to a real course:
      "You just built a DIGITAL FTE."
      "-> AgentFactory Course 15: From Agent to Digital FTE"
   A small icon row shows the 4 parts the player assembled,
   labeled with their real names: Brain, Hands, Training, Shift.
```

- **Juice:** the four part-icons from Scene 2 slide in with their real-world labels - tying the game metaphor directly to the real concept.
- **Audio:** a soft "aha" chime.
- **Concept: serious fun** - the play has real-world meaning and a concrete next step in the actual curriculum.

Frame 3 — The Two Choices

```
   Two buttons, equal weight:
        [ Maybe later ]            [ Show me > ]
   - Maybe later  -> card slides away, straight to Level 2
   - Show me      -> opens the AgentFactory course link in-app
                     (a gentle web view), then returns to the game
Nova:  "No rush - it'll be here whenever you're curious."
```

- **Juice:** both buttons are friendly and equal (no dark-pattern nudging); choosing either feels good. "Show me" opens a lightweight in-app view; "Maybe later" dismisses with a soft swoosh.
- **Audio:** a light confirm tap either way.
- **Concept:** respectful, opt-in learning funnel - turns enjoyment into genuine curiosity about AgentFactory's real courses.

Frame 4 — Return to Play

```
Either choice lands the player at the Level 2 entrance:
     "WORLD 1 - The Startup Garage  |  Level 2: The Magic Words"
     (o_o)  <- Nova: "On to the next hire!"
```

- **Transition:** smooth return to the World 1 map / Level 2 intro, keeping momentum.
- **Concept:** the loop continues - fun first, real learning always one tap away.

Designer Notes for These Two Screens

- **Sequence the rewards (Scene 6):** never dump all rewards at once - coins, then XP, then the unlock, each as its own little hit. This stretches the satisfaction and teaches what each reward means.
- **The unlock is the lesson (Scene 6):** Night Shift is not just a reward - it is the literal demonstration of "a Digital FTE works 24/7." Make the ticking desk-clock visible and ongoing.
- **The bridge must never nag (Scene 7):** keep it a small, dismissible corner card with two equal choices. A learning funnel that feels like an ad breaks trust; one that feels like a friendly tip builds it.
- **Tie the metaphor to the real names (Scene 7):** showing Brain/Hands/Training/Shift with their real labels is the single highest-value teaching moment - it converts a game mechanic into a transferable concept.
- **Consistent reward grammar:** the same coin "cha-ching," XP tick, and unlock chime used here match every other win in the game.
- **Reduced-motion mode:** confetti and card-flips become soft fades; all audio cues and timings preserved.

Part 9 — Character Bio: Nova, Your Guide

This section fully specifies Nova so an art, animation, and writing team can build her consistently. Nova is the player's companion and tutor across every scene; her job is to make a complex subject (Agentic AI) feel warm, easy, and fun.

At a Glance

Field	Detail
Name	Nova
Role	Guide mascot, tutor, hype-friend
Species	A small, sentient orb of light (a friendly "spark of intelligence")
Pronouns	She / her

Field	Detail
First appears	Level 1, Scene 0
Core job	Teach by encouraging, never by lecturing
One-line essence	“Half mentor, half best friend — the spark that believes in you.”

Visual Design

- **Form:** a floating orb roughly the size of the player’s worker’s head, hovering with a gentle bob.
- **Color:** glows in the brand palette — a violet core (#6D28D9) with a teal aura (#14B8A6); brightens to gold (#FBBF24) when celebrating.
- **Face:** two large, expressive eyes (no mouth needed — emotion reads through the eyes and brow). She can squint, widen, sparkle, and do a happy “upturned” smile-eyes.
- **Signature feature:** a wispy “spark tail” that trails behind her movement and flicks when she’s excited.
- **Scale & framing:** she lives in the corner or beside the worker, never blocking the play area; she pokes in from the side rather than covering the screen.
- **Silhouette test:** instantly recognizable as a glowing dot with eyes and a spark tail — clean at any resolution (matches the low-poly art direction).

Personality

- **Upbeat & encouraging:** every player action is “great,” “bold,” or “crystal clear.” She finds the good even in a silly mistake.
- **A little cheeky:** she teases gently (the “bold choice” line on a silly spec) but never mocks.
- **Concise:** she speaks one short line at a time — she trusts the player to learn by doing.
- **Calm under chaos:** when the worker melts down, Nova laughs with the player, not at them, and reframes it as a lesson.
- **Genuinely on your side:** she frames everything as “we” and “your worker,” reinforcing that the player is the boss.

Voice & Tone Rules (for writers)

- **Short.** One sentence per beat. If it needs two, cut it to one.
- **Second person, warm.** “You set it up great!” not “The setup is complete.”
- **Show, don’t lecture.** Point to the next action; let consequence teach the lesson.
- **Celebrate specifics.** “Crystal clear!” beats “Good job” — name what they did well.
- **Never punish.** Failure lines are playful and forward-looking (“Let’s give it a REAL spec”).

- **No jargon up front.** Real terms (Digital FTE, spec, MCP) are introduced only after the player has felt the concept.
- **Earned hints.** Nova only nudges with the answer after a player struggles — capable players keep the challenge.

Sample Catchphrases

- Greeting: “Welcome, boss!”
- Encouragement: “You set it up great!” / “Crystal clear!”
- Gentle tease (on a silly choice): “Hmm... bold choice.”
- Reframing a mistake: “HA! Fuzzy orders = chaos. Let’s fix it together.”
- The golden rule: “Humans set intent. Agents execute. Humans verify.”
- Big win: “Boom! Happy customer, happy boss.”
- The Digital FTE magic: “This one never sleeps.”
- Sign-off into the next level: “On to the next hire!”

Animation Set (for the animator)

State	What Nova does
Idle	Gentle bob and slow blink; spark tail drifts
Talking	Slight forward lean, eyes animate, tail flicks on emphasis
Excited / celebrating	Spins, brightens to gold, tail sparkles, tiny confetti
Teasing	One eye squints, a playful little wobble
Thinking / hinting	Eyes drift up, a small “?” or lightbulb spark above her
Reassuring (after a fail)	Softens color, leans in close, warm slow blink
Peeking in (Scene 7)	Slides in from a screen edge, only head/eyes visible

Accessibility for Nova

- **Captions:** every spoken line is fully captioned; Nova never communicates critical info by voice alone.
- **Color-independent emotion:** her feelings read through eye shape and motion, not just glow color (colorblind-safe).
- **Reduced motion:** spins and tail-sparkles become soft fades; her timing and lines are preserved.
- **Skippable:** her lines can be tapped-through; she never blocks or slows an experienced player.
- **Readable text:** caption text uses the UI’s adjustable size and high-contrast settings.

Why Nova Works

Nova turns a potentially intimidating topic into a friendly conversation. By being concise, encouraging, and consequence-driven, she embodies the game’s three core promises —

easy (one short nudge at a time), **fun** (she celebrates and teases), and **interactive** (she reacts live to what the player does). She is the emotional through-line that makes every scene feel personal.

Part 10 — The 2-Week MVP Build Plan

This is a developer-facing plan to build a playable vertical slice of **Level 1** in two weeks. The goal is not a finished game — it is to **prove the core loop is fun**: assemble a worker, write a spec, press GO, verify, get rewarded. If playtesters smile at that loop, the concept is validated.

Scope: What the MVP Includes (and Excludes)

In scope (the vertical slice):

- One full playthrough of Level 1, Scenes 0 through 6 (Scene 7 bridge optional)
- The 4-part drag-and-drop Worker Builder (Scene 2)
- The Spec Card with the "silly spec" gag and the correct path (Scene 3)
- The live work animation (Scene 4) and Verify/Approve (Scene 5)
- Rewards: coins, XP, and the Night Shift unlock (Scene 6)
- Nova as a guide (static art + text lines; full animation later)
- Core "juice": snap feedback, sparkle, click-ding, coin arc, confetti

Out of scope (deliberately deferred):

- Levels 2 through 26, the world map, the Mode fork
- The idle/factory-management layer beyond a single ticking clock
- Leaderboards, sharing, accounts, monetization
- Full Nova animation set, voice-over, multiple worker types
- Cloud save (local save only)

Team Assumption

A small team of 2 to 3: one engineer (Unity/C#), one artist (2.5D/UI), and a shared design/audio role. The plan also works for a solo developer by treating each "role-day" as sequential.

Technical Setup (Day 0 / pre-flight)

- Unity 6 LTS project with the Universal Render Pipeline (URP)
- DOTween for tween/juice; TextMeshPro for UI text
- Target builds: WebGL (primary, for no-install demos) and Android

- Source control (Git + Git LFS for art assets)
- A grey-box art kit (Kenney/Synty placeholders) so engineering never waits on final art

Week 1 — Build the Core Loop (grey-box, fun-first)

Day	Focus	Deliverable
1	Project + scene skeleton	Unity project builds to WebGL; empty Level 1 scene with placeholder UI zones (header, worker stage, Nova bar, thumb zone)
2	Worker Builder mechanic	Drag-and-drop 4 parts into slots with magnetic snap; progress dots fill; grey-box art
3	Builder juice + Nova hints	Snap spring-overshoot, sparkle particles, click-ding SFX; Nova one-line hints drive the sequence; worker “gains eyes” on Brain
4	Spec Card mechanic	Tap-to-fill word-tiles into 3 blanks; validation (correct vs. silly); undo by tapping a filled blank
5	The “silly spec” gag + correct path	Slapstick fail animation + sad-trombone (no penalty, auto-reset); green “SPEC LOCKED IN” on the correct spec

End of Week 1 milestone: a player can build a worker and write a valid spec, with satisfying feedback. Internal smoke-test the build → spec flow.

Week 2 — Close the Loop, Polish, Playtest

Day	Focus	Deliverable
6	Press GO + work animation	GO button juice; sequential tool-light-up (Read → Check Order → Policy → Write); progress bar; typewriter reply
7	Verify / Approve	Verify panel; Approve and Improve buttons; customer reacts (heart + 5 stars); coin arc on approve
8	Rewards + Night Shift unlock	Sequenced reward tally (coins → XP → unlock); XP bar level-up; ticking desk-clock for the idle hook
9	Art pass	Replace grey-box with first real 2.5D assets: worker, Nova, customer, UI skin, brand palette, lighting + bloom
10	Audio + juice pass	Full SFX set (click-ding, coin cha-ching, star chimes, power-up, sad-trombone), music loop, confetti/screen-shake tuning

Day	Focus	Deliverable
11	Accessibility + reduced-motion	Colorblind-safe checks, caption text, reduced-motion toggle, big tap targets verified
12	Build, bug-fix, instrument	Stable WebGL + Android builds; add lightweight analytics (loop completion, retries, drop-off points)
13	Playtest	5 to 8 fresh players; observe Level 1 completion, where they hesitate, and whether they smile at the core loop
14	Iterate + report	Fix the top 3 friction points found in testing; write a short validation report against the success metrics

End of Week 2 milestone: a polished, playable Level 1 vertical slice with real art, audio, and instrumentation — ready to demo and to greenlight (or not) full production.

Definition of Done (MVP)

- A new player can complete Level 1 unaided, start to finish
- Every core interaction has its feedback (snap, sparkle, ding, coin arc, confetti)
- Runs at a stable 60fps on a mid-range Android phone and in-browser (WebGL)
- Reduced-motion and caption settings function
- Analytics capture loop-completion and drop-off

Success Metrics (the validation bar)

Pillar	Measured by	Target
Easy	Level 1 completion rate (fresh players)	> 90% finish unaided
Fun	"Was that fun?" + observed smiles/replays	Majority yes; spontaneous replay
Interactive	Time-to-first-action; actions per minute	Fast first action; low confusion time
Graphics/feel	Frame rate + qualitative delight	Stable 60fps; testers react to the juice
Learning	Post-play: name the 4 FTE parts	> 80% recall

If these targets are met, the assemble → spec → verify loop is proven and the project is ready to scale to the full 26-level game. If not, the report pinpoints exactly which pillar needs rework before further investment.

Key Risks and Mitigations

- **Risk: the core loop is not fun.** Mitigation: build it grey-box in Week 1 and smoke-test early — do not wait for art to learn this.
- **Risk: art slips the schedule.** Mitigation: ship on placeholder assets; the art pass is Day 9, after the loop is proven.
- **Risk: the spec lesson does not land.** Mitigation: the "silly spec" gag is the teaching device — playtest it specifically and tune the comedy and reset feel.
- **Risk: WebGL performance.** Mitigation: target low draw calls and atlased textures from Day 1; test in-browser every build, not just in the editor.

Part 11 — GitHub Sync & Deployment

The playable prototype and this design document are version-controlled on GitHub and published as a live, in-browser game via GitHub Pages.

Live Links

Resource	URL
Play the game (GitHub Pages)	https://caats-kkhan.github.io/agent-factory-tycoon/
Source repository	https://github.com/CAATs-kkhan/agent-factory-tycoon

The repository contains the playable `index.html`, this design document (`.md`, `.docx`, `.pdf`), and the `README.md`.

One-Time Setup (how it was synced)

Prerequisites: Git and the GitHub CLI (gh) installed, and `gh auth login` completed.

1. **Initialize the repo** (run from inside the game folder):

```
git init
git add -A
git commit -m "Agent Factory Tycoon: playable prototype + design docs"
git branch -M main
```

2. **Create the GitHub repo and push** (the GitHub CLI creates it and pushes in one step):

```
gh repo create agent-factory-tycoon --public --source=. --remote=origin
 --push
```

3. **Enable GitHub Pages** (serves the game from the `main` branch root):

```
gh api -X POST repos/CAATs-kkhan/agent-factory-tycoon/pages \
  -f "source[branch]=main" -f "source[path]=/"
```

   After about 1–2 minutes the game is live at the Pages URL above. Verify the build with:

```
gh api repos/CAATs-kkhan/agent-factory-tycoon/pages/builds/latest --jq .
status
```

Update Workflow (for every future change)

After editing `index.html` (or any file), publish the change:

```
git add -A
git commit -m "describe your change"
git push
```

GitHub Pages automatically rebuilds within about 1–2 minutes. Hard-refresh the live page (`Ctrl + F5`) to bypass the browser cache and see the update.

Regenerating the Documents

If the design document Markdown changes, rebuild the `.docx` and `.pdf` (see the README for the exact commands), then commit and push them alongside the source so the repository stays in sync.

Sources

- The Agent Factory Thesis — https://agentfactory.panaversity.org/docs/thesis
- Synthesis: The Digital FTE Vision — https://agentfactory.panaversity.org/docs/General-Agents-Foundations/agent-factory-paradigm/synthesis-digital-fte-vision
- Project / Simulation Labs — https://agentfactory.panaversity.org/docs/simulation-labs
- Getting Started: Crash Courses — https://agentfactory.panaversity.org/docs/getting-started
- About the AI Agent Factory — https://agentfactory.panaversity.org/docs/about
- GitHub: panaversity/agentfactory — https://github.com/panaversity/agentfactory